Chapter 1
Ray Picture of light

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Activity (1-1) some simple and helpful videos to understand the nature phenomena of light ,the interaction of light with the different surfaces and the shadow.

1-What is Light?

???? ?????? - YouTube.MP4

???? ?????? - YouTube_2.MP4

2-The different Kinds of Surfaces.

Transparent, Opaque and Translucent objects - C8SE6 - YouTube.MKV

3-The Shadow.

????? ???? ????? ?????? ??????? - YouTube.MKV

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lecture (1)

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1-1- Introduction.

Optics is the branch of physics that studies the behavior and properties of light, including its interactions with matter and the construction of instruments that use or detect it. Optics usually describes the behavior of visible, ultraviolet, and infrared light. Because light is an electromagnetic wave, other forms of electromagnetic radiation such as X-rays, microwaves, and radio waves exhibit similar properties.
Most optical phenomena can be accounted for using the classical electromagnetic description of light. Complete electromagnetic descriptions of light are, however, often difficult to apply in practice. Practical optics is usually done using simplified models:
Geometric optics, treats light as a collection of rays that travel in straight lines and bend when they pass through or reflect from surfaces.
Physical optics is a more comprehensive model of light, which includes wave effects such as diffraction and interference that cannot be accounted for in geometric optics.
Optical science is relevant to and studied in many related disciplines including astronomy, various engineering fields, photography, and medicine (particularly ophthalmology and optometry).
Practical applications of optics are found in a variety of technologies and everyday objects, including mirrors, lenses, telescopes, microscopes, lasers, and fiber optics.

Now it is very important to know , something about light and its different sources as an introduction to study ,its interaction with matter.

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Source of light

Industrial sources

Torches

old lamp

Candles

Natural sources

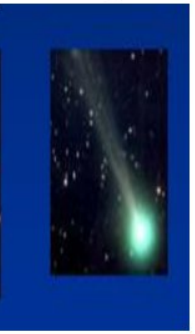
Moon-

sun-

lightning

Fire

Falling star



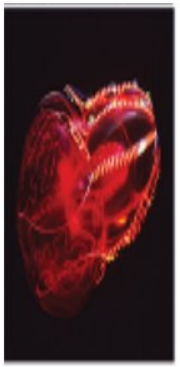
Non-self-lighting

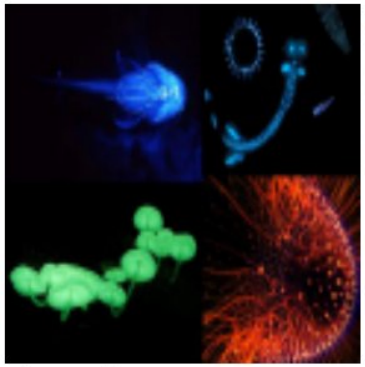
Shower

Infrared image

Optical fiber

Self-lighting





Coral reefs

Tywan tear

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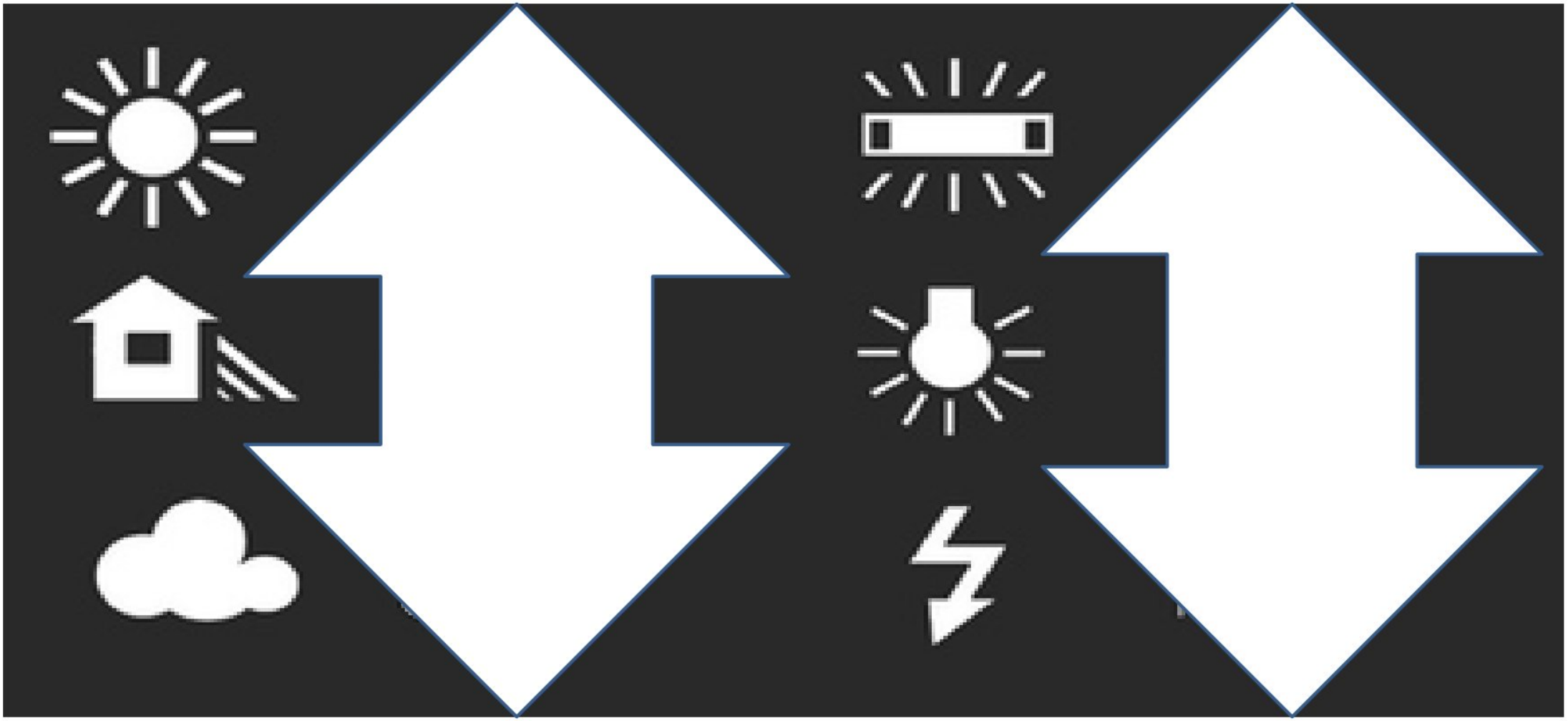
Activity (1-2)All theses Signs you must know. General discussion.



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1-1-a- Nature of light:

Historical information:

The Greeks believed that light consisted of tiny particles(corpuscles) that were emitted by light source and these particles stimulated the perception of vision upon

b)

Striking the observer's eye.

Fig.(1-1 –a, b)

1-Newton (at the 17 century) used this theory to explain the reflection and the refraction phenomena of light. He proposed that these particles have the following properties:

a-They propagate in straight lines .

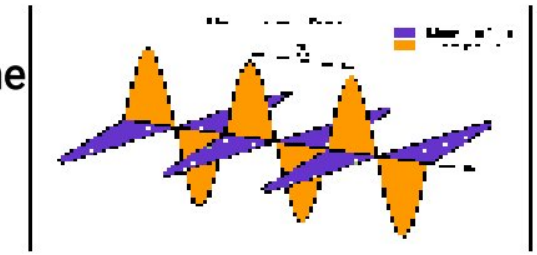
b-At the surface which separating two different media, they are exposed to two short- range attractive and repulsive forces which support the reflection and the refraction phenomena of light ,as shown in Fig.(1-1-a, b)

c-The particles emitted from the light source are stimulated the sense of sight upon entering the eye.

The disadvantage of Newon,s theory:

He could not explain refraction phenomena , because the speed of light in water higher than in air (this against the experimental observations).

2-The astronomer Christian Huygens showed that a wave theory of light could explain the refraction phenomena of light.



3- Thomas young (In 1801):

He provided and applied the wave nature of light.

4-Maxwell (1873):

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Maxwell calculated the speed of these waves by his equation: $C = \frac{1}{\sqrt{\frac{\epsilon_0}{\mu_0}}}$ =3x10^8 m/s

Where ϵ_0 is the electric permittivity and μ_0 is the magnetic permeability of space.

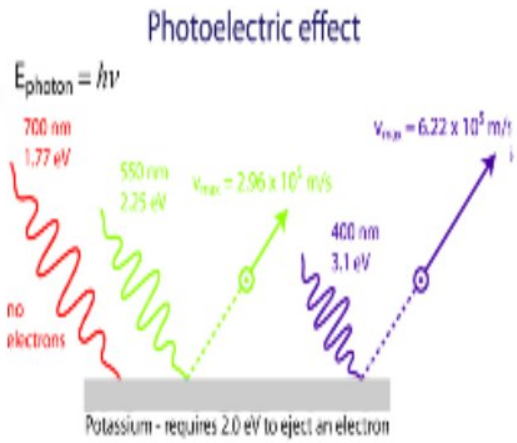

Fig. (1-3)

5- Hertz (1887):

Hertz proved experimentally that these waves exhibit propagation characteristics similar to those of light; reflected ,refracted ,polarized, etc.

**Although the wave model and the classical theory of electricity and magnetism were able to explain most known prosperities of light , they could not explain some experiments. The most striking of these is the photoelectric effect(Fig.1-3).

6- Max Plank (1900):

He discover the quantization model of light as shown in Fig.(1-4).

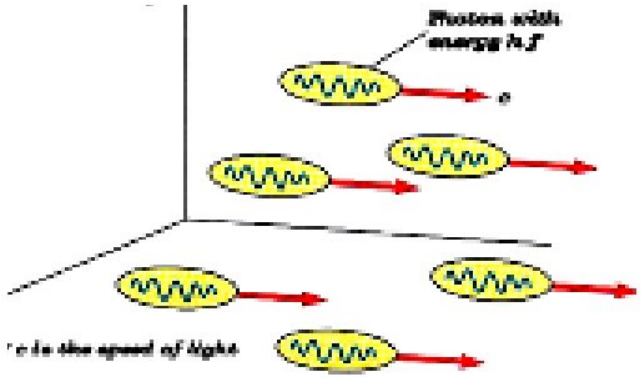

Fig.(1-4)

7-Einstein (1905):

The quantization model of light, which assumes that the energy of light wave is present is present in particles called Photons ,hence the energy is said is be quantized. He put his equation: **E=hf** where f is the frequency of the electromagnetic waves, and h is Planks constant : h = $6.63\ x10^{-34}$ J/s.

After all of these experiments from numerous scientists ,light must be regarded as having a dual Nature :

(Light exhibit the characteristics of a wave in some situations and of a particle in other situations).

So the question :" is light a wave or a particle?"

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1-2- Huygens Principle: (wave theory of light) Huygens Principle is a geometric construction:

"All points on a given wave front are taken as point sources for the production of spherical Secondary waves, called wavelets which propagate outward through he medium with speed characteristic of waves in that medium – after some time interval has passed, the new position of the wave front is the surface tangent to the wavelets" , as shown in Fig.(1-5 –a , b , c).

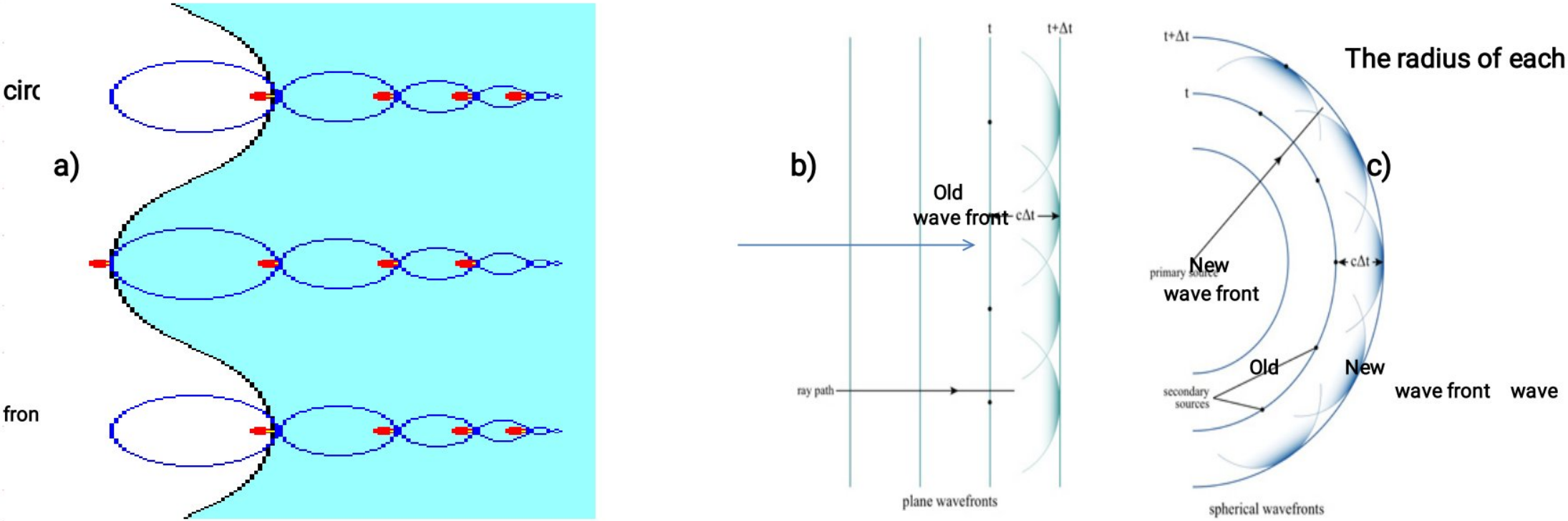

Fig.(1-5 –a , b , c)

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1-3.The Electromagnetic spectrum: (as shown inFig. (1-6)and application in Fig.(1-7)

Light ,as Maxwell pointed out, is a member of the electromagnetic wave family . The spectrum of these waves covers the wide range of wavelengths($10^{-14} \sim 10^{14}$ m) , which corresponds to the frequency range($10^{2} \sim 10^{23}$ Hz) .
These waves have electromagnetic nature and propagate in free space with speed of light C.

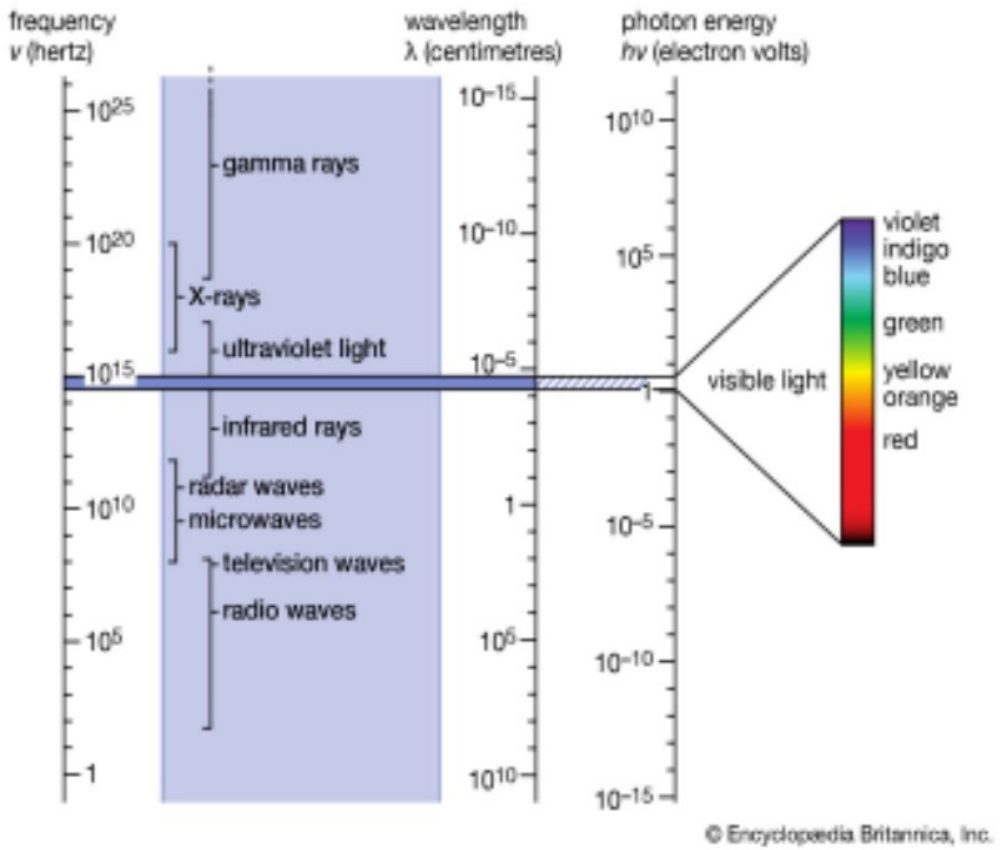

Fig.(1-6)The Electromagnetic

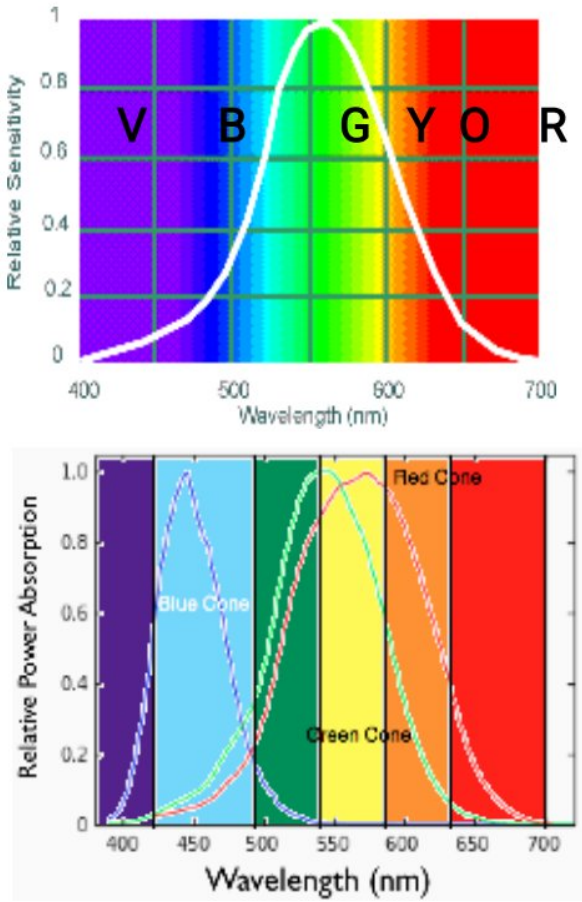

Fig. (1-7)Sensitivity spectrum of a standard

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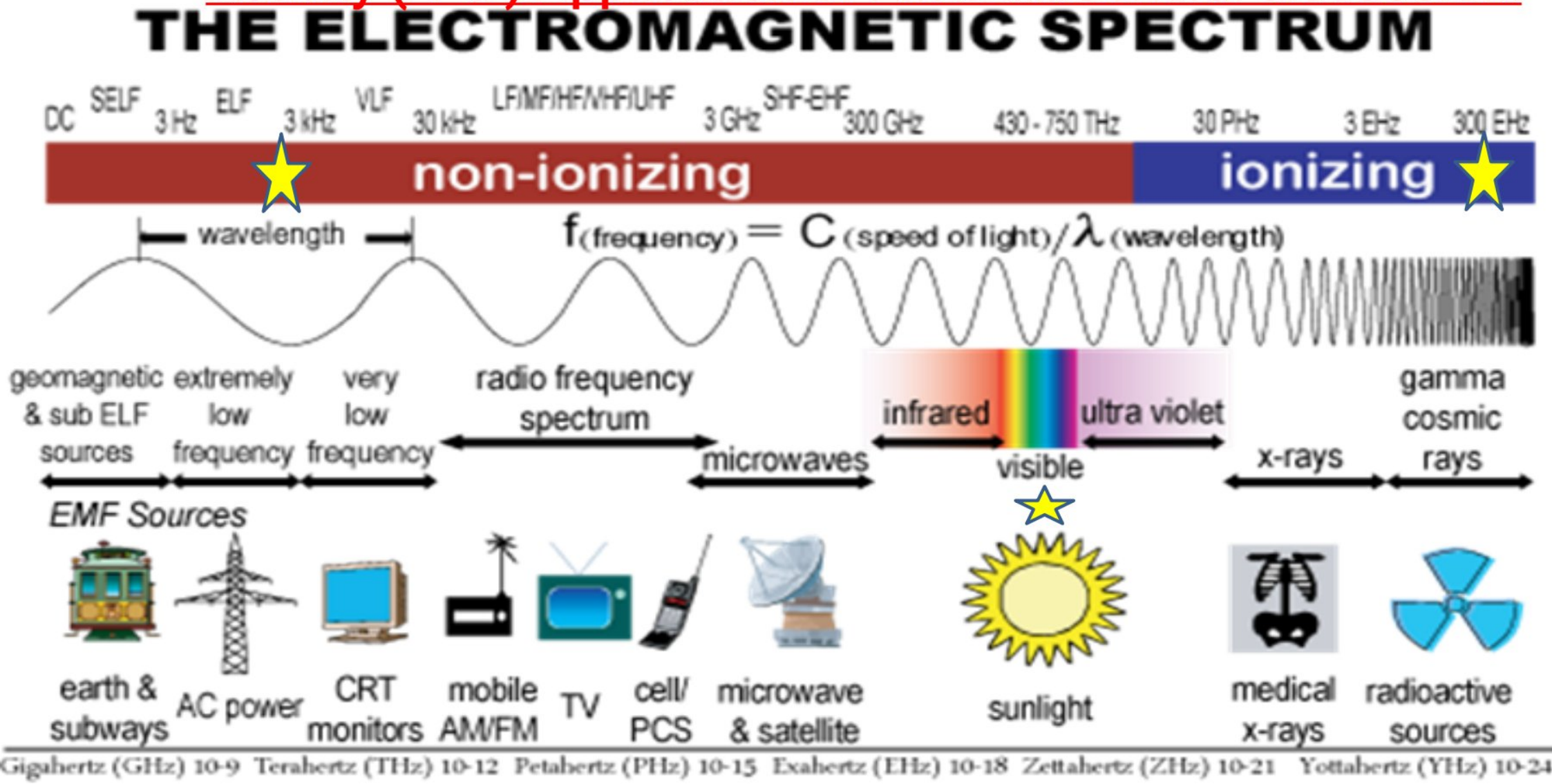
Activity (1-3) Application of EM S .....General discussion.



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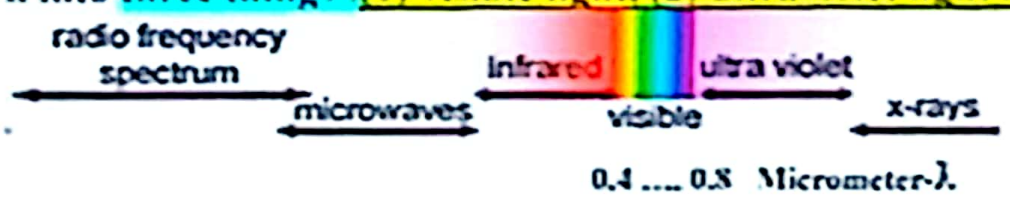
The sun core Nuclear Fusion reaction: ^{1}H change under ..to ^{2}He + Huge Energy → Light

(+ Pressure +high temperature) (heat + Radiation)

Ordinarily, sunlight: is broken down into three things :(1) visible light, (2) ultraviolet light and (3) infrared radiation.



Light is the principal means .... we can transmit and receive information..... to and from objects around us and throughout the Universe.

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Polarimeter –Steps of Invention-From manual to digital, Analysis of sugar (plant and every things)

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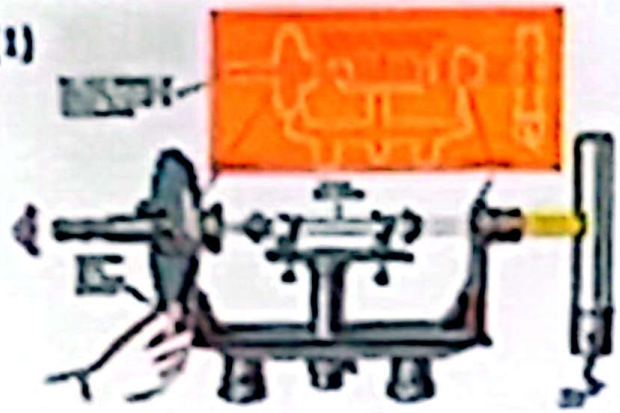

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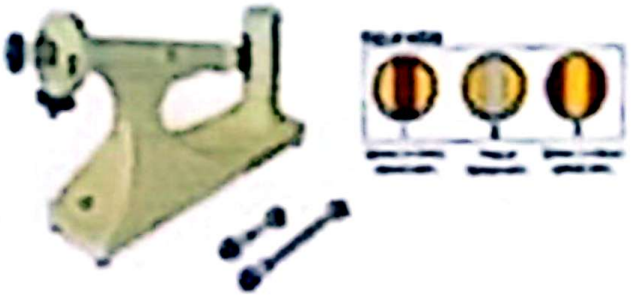

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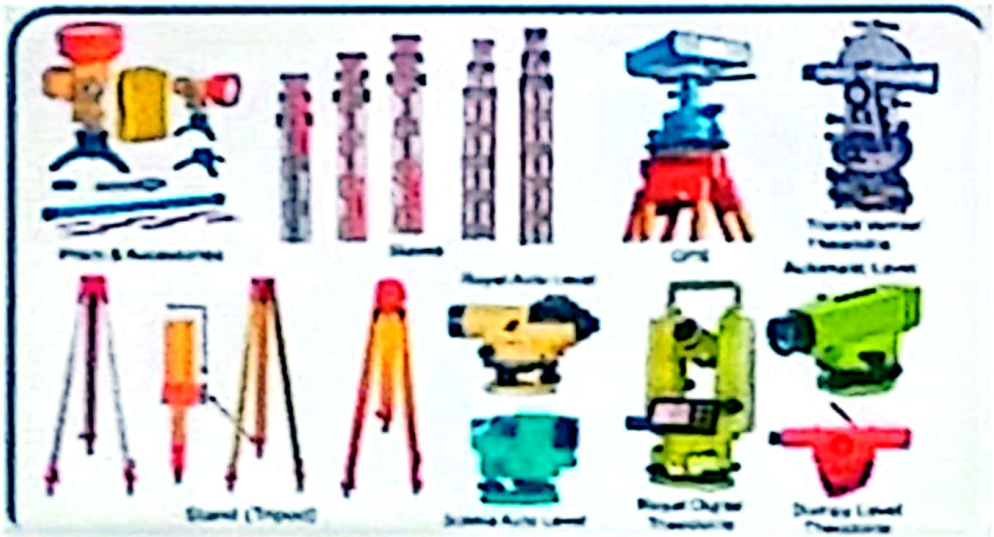

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